

Business Questions

1. Which laptop company has the highest average price?

Answer: Razer has the highest average laptop price in the dataset. On average, Razer laptops are more expensive than laptops from other companies.

2. Does RAM affect laptop price?

Answer: Yes. Laptop price generally increases as RAM capacity increases. Laptops with higher RAM (such as 16 GB, 32 GB, or 64 GB) have a much higher average price than laptops with lower RAM (such as 4 GB or 8 GB).

3. Does SSD storage increase laptop price?

Answer: Yes. Laptops with SSD storage have a higher average price than laptops without SSD storage. SSD provides better performance, which contributes to the higher price.

4. Which laptop type is the most expensive?

Answer: Workstation laptops are the most expensive type in the dataset. They have the highest average price compared to Gaming, Ultrabook, Notebook, Netbook, and 2-in-1 Convertible laptops.

5. Does processor type affect laptop price?

Answer: Yes. Processor type has a significant impact on laptop price. Laptops with Intel Core i7 processors have the highest average price, while laptops with Intel Core i3 or AMD processors are generally less expensive.

Conclusion

The Exploratory Data Analysis (EDA) of the laptop price dataset shows that **RAM, processor type, storage type (SSD), and laptop brand have a significant impact on laptop prices**. Data cleaning and outlier handling improved the quality of the dataset, while feature engineering and encoding prepared it for machine learning. Overall, the dataset provides valuable insights into the factors that influence laptop prices and is suitable for predictive modeling.